

10.1.2 . Reverse a String

A] Algorithm

Step 1: Start

Step 2: Read input string from the user and store it in user_input

Step 3: Define a function reverse_string(input_string)

Step 4: Inside the function, initialize an empty string re = ""

Step 5: Find the length l of input_string

Step 6: Loop from index l-1 down to 0 (reverse order):

Step 6a: Append the character at current index to re

Step 7: After the loop, return re (the reversed string)

Step 8: Call reverse_string(user_input) and store the result in result

Step 9: Print the original string user_input

Step 10: Print the reversed string result

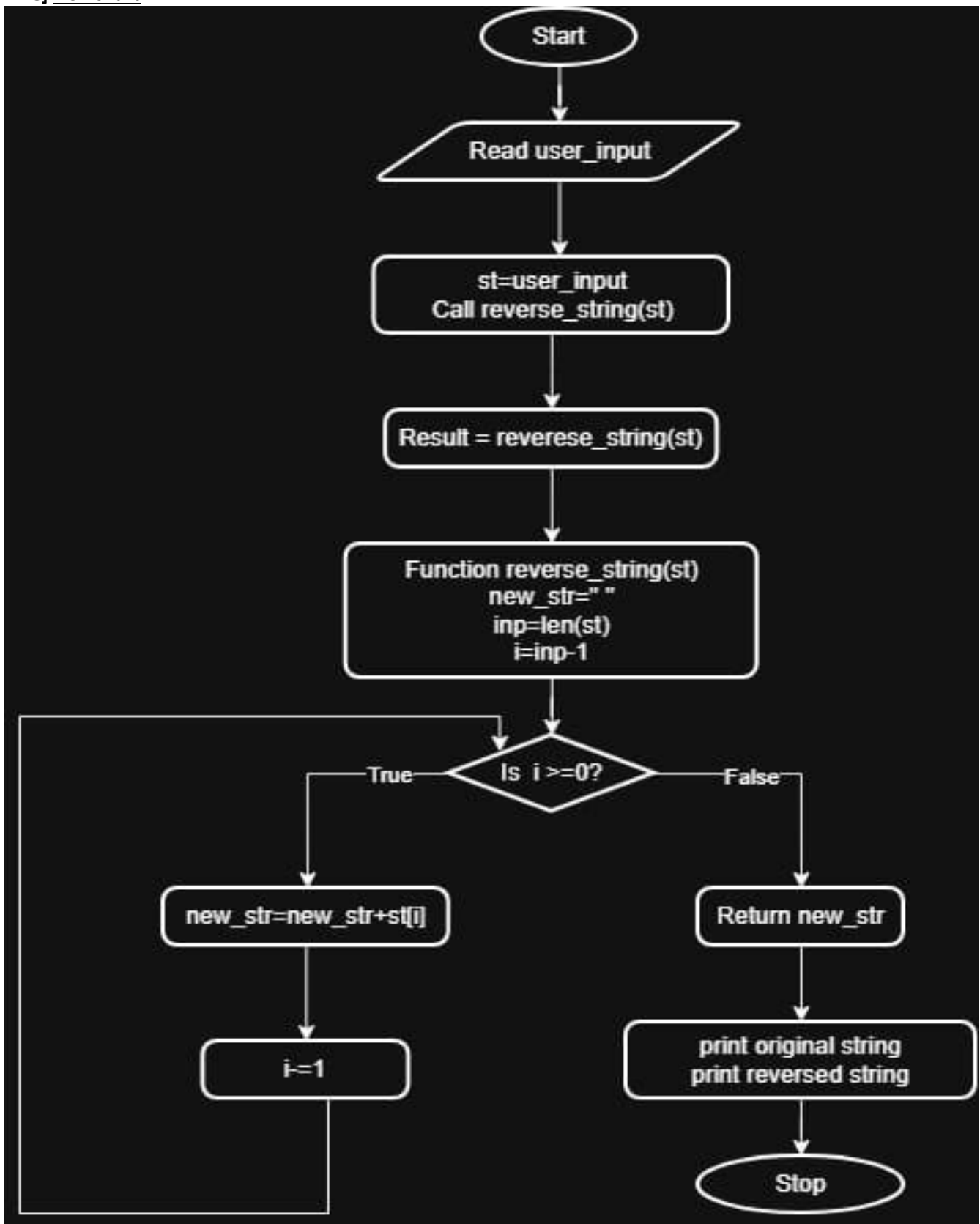
Step 11: End

B] code

```
def reverse_string(input_string):  
    re=""  
    l = len(user_input)  
    for i in range(l-1,-1,-1):  
        re += user_input[i]  
    return re
```

```
user_input = input("Enter a string: ")  
result = reverse_string(user_input)  
print(f"Original String: {user_input}")  
print(f"Reversed String: {result}")
```

C) Flowchart



10.1.2. Reverse a String

00:02 A L [?] -

Write a Python program that prints the reverse of a given input string. The program uses a `reverse_string()` function that takes a string as input and returns its reversed string.

Input Format:

- The program prompts "Enter a string:" and reads a string representing the string to be reversed.

Output Format:

- The first line prints the original string, in the format:

Original String: <input_string>

- The next line prints the reversed string, in the format:

Reversed String: <reversed_string>

Sample Test Cases

+

ReverseS...

Submit

```
1 def reverse_string(s):
2     # Write the code...
3     return s[::-1]
4 user_input = input("Enter a string: ")
5 result = reverse_string(user_input)
6 print(f"Original String: {user_input}")
7 print(f"Reversed String: {result}")
```

Explorer

Debugger

Average time

0.007 s

Maximum time

0.009 s

6.75 ms

9.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 9ms Debug

Expected output

Actual output

Enter a string: PythonProgramming

Enter a string: PythonProgramming

Original String: PythonProgramming

Original String: PythonProgramming

Reversed String: gniimargorPnohtyP

Reversed String: gniimargorPnohtyP

Test case 2 5ms

Terminal Test cases

< Prev Reset Submit Next >

D] OUTPUT